Akshay Patil

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EDUCATION

Georgia Institute of Technology – Atlanta, GA **August 2026 – May 2029**
Bachelor of Science in Aerospace Engineering
Current coursework: CS 1371 Computing, MATH 2551 Multivariable Calculus, COE 2001 Statics, ME 1670 Graphics, AE 1601 Aerospace Engineering, MSE 2001 Materials

School of Science and Engineering at Townview – Dallas, TX **August 2022 – May 2026**
3.97 Unweighted GPA | Rank 2 | 1600 SAT | National Merit Scholar | NASA High School Aerospace Scholar

RESEARCH

IVALab, Georgia Tech | Undergraduate Researcher **September 2026 – Present**

- Developing robot navigation algorithms for pedestrian dense environments, mixing learned and classical planners and benchmarking them in the Arena simulator

Hypersonics VIP, Georgia Tech | Data and Communications Team **August 2026 – Present**

- Building a sensor package to measure pressure, temperature, and airflow inside a sonic fatigue test facility
- Developing parabolic mirrors to generate a uniform sound field across the facility's test section

EXPERIENCE

Hardpoint Robotics | Co-Founder **June 2026 – Present**

- Designing NDAA-compliant development UAVs, owning the airframe: CAD, 3D printed CF-Nylon parts, and carbon fiber structures made with outside vendors
- Designing around sourcing as a hard constraint, since every component has to be compliant and in stock

Salvo Dynamics | Design Engineer **August 2026 – Present**

- Turning the aero body of a Mach 1 Challenge RC jet into a load-bearing structure in CAD, and laying out the avionics that go inside it
- Designing a 3D printed airframe for testing, plus the assembly jigs used to build the interceptor jets

SWYFT Robotics | Engineering Design Intern **May 2025 – May 2026**

- Designed sprockets, gears, and pulleys for FIRST Tech Challenge and FIRST Robotics Competition teams
- Designed a modular robot cart for FIRST Robotics Competition teams with a detachable programming station

PROJECTS

Autonomous House-Mapping Robot | Personal Project **2025 – Present**

- Jetson Orin Nano robot that explores a house with no operator and counts windows, using an MPPI controller, RTAB-Map visual SLAM, and frontier exploration I wrote
- No lidar: obstacles come from monocular depth classified against a fitted ground plane; windows from three open-vocabulary detectors filtered by a vision-language model
- Holds a commanded goal to within 7 cm behind a 50 Hz fail-toward-stop safety governor

Pickleball Ball Boy UAV | Personal Project **May 2025 – Present**

- Benchmarked five ball detection and four localization methods; AprilTag localization was the most accurate by a wide margin
- Split the drag model into horizontal and vertical coefficients, bringing landing prediction inside a 20 cm zone

ENGINEERING TEAMS

FIRST Tech Challenge Team 7172 | Captain and Lead Designer **September 2021 – May 2026**

- Lead designer on four robots; 2026 World Championship Finalist Alliance, 19 event wins and 63 awards over five seasons
- Designed a shooter that sorts and fires three balls in 0.2 to 0.5 s using a wedge spindexer and a flywheel on a 200 Hz servo turret

Georgia Tech Experimental Rocketry | Externals Team **August 2026 – Present**

- Designing a fin mounting jig that has to locate fins on an already wrapped carbon fiber tube whose exact dimensions are no longer known, and hold fin cant to within a tenth of a degree

Yellow Jacket Space Program | Turbomachinery, Engine Development **August 2026 – Present**

- Designing electric pumps and turbopumps so the vehicle can drop the large pressurant tanks a pressure-fed design needs

SKILLS

Robotics: ROS 2 (Nav2), RTAB-Map visual SLAM, MPPI control, AprilTag localization
Programming: Python (NumPy, Pandas, TensorFlow), MATLAB, JavaScript, SQL, Git
Hardware: NVIDIA Jetson, Raspberry Pi, Arduino, REV Control Hub
CAD: SolidWorks, Fusion 360, Onshape, Autodesk Inventor, 3D printing
Certifications: FAA Part 107 Remote Pilot, Autodesk Certified User (Inventor)